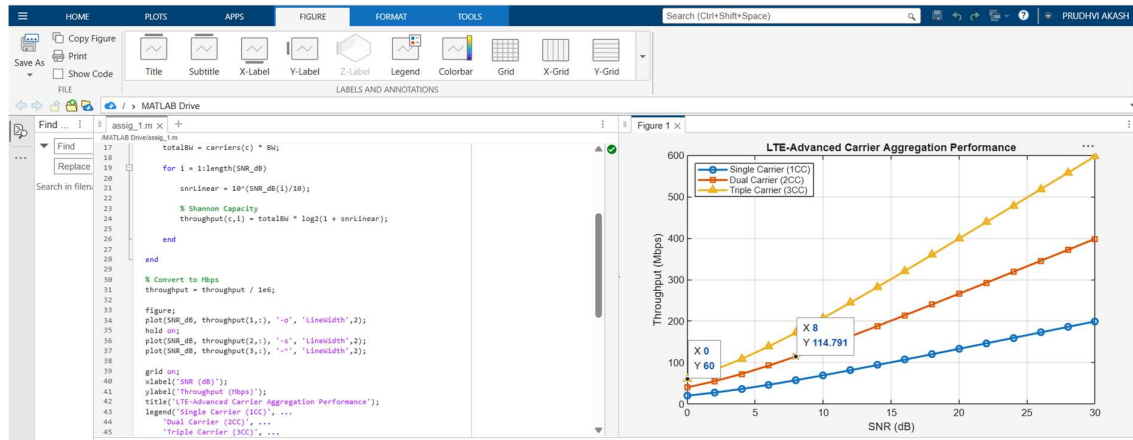




```
clc;
```

```
clear;
```

```
close all
```

```
% LTE Carrier Aggregation Throughput Simulation
```

```
SNR_dB = 0:2:30;    % SNR values in dB
```

```
BW = 20e6;          % Bandwidth of one carrier (20 MHz)
```

```
% Number of aggregated carriers
```

```
carriers = [1 2 3];
```

```
throughput = zeros(length(carriers), length(SNR_dB));
```

```
for c = 1:length(carriers)
```

```
    totalBW = carriers(c) * BW;
```

```
    for i = 1:length(SNR_dB)
```

```
        snrLinear = 10^(SNR_dB(i)/10);
```

```
        % Shannon Capacity
```

```
        throughput(c,i) = totalBW * log2(1 + snrLinear);
```

```
    end
```

```
end
```

```
% Convert to Mbps

throughput = throughput / 1e6;

figure;

plot(SNR_dB, throughput(1,:), '-o', 'LineWidth',2);

hold on;

plot(SNR_dB, throughput(2,:), '-s', 'LineWidth',2);

plot(SNR_dB, throughput(3,:), '-^', 'LineWidth',2);

grid on;

xlabel('SNR (dB)');

ylabel('Throughput (Mbps)');

title('LTE-Advanced Carrier Aggregation Performance');

legend('Single Carrier (1CC)', ...

'Dual Carrier (2CC)', ...

'Triple Carrier (3CC)', ...

'Location','northwest');
```